YouTube Data Analytics and Sentiment Analysis

Final Project Report

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Introduction

In this project, we used the YouTube Data API and Python to analyze the 3Blue1Brown YouTube channel. YouTube analytics is helpful as it allows to assess the performance of the content based on the number of views, comments, patterns of publication, and interaction with the audience. Sentiment analysis can also be significant since it will add sense to those numbers as it will indicate whether titles and comments are predominantly positive, neutral, or negative. Combined, these techniques assist us in comprehending channel performance as well as viewer response.

This project aimed to retrieve video metadata, determine trends of views by time, common topics in video titles, frequency of publication, and sentiment analysis of titles and comments. The project is applicable to the integration of big data since we gathered data via an external API, cleaned and structured it into datasets, and used visualization and analysis tools to extract insights. Overall, the project demonstrates that raw social media data can be converted into valuable insights into the content strategy, audience engagement, and patterns of online discussions.

Methodology and Results Discussion

Data Extraction and Preliminary Analysis

We were able to extract a structured dataset of 3Blue1Brown channel. The last dataset consisted of 232 videos and had the necessary fields: video ID, title, publish time, view count, and comment count. This ensured that our extraction procedure was successful and provided us with a full dataset to continue with the rest of the project. The channel query indicated a total number of channels video of 234, and the extracted data retrieved 232 video records, which were still sufficient to conduct the necessary analysis activities.

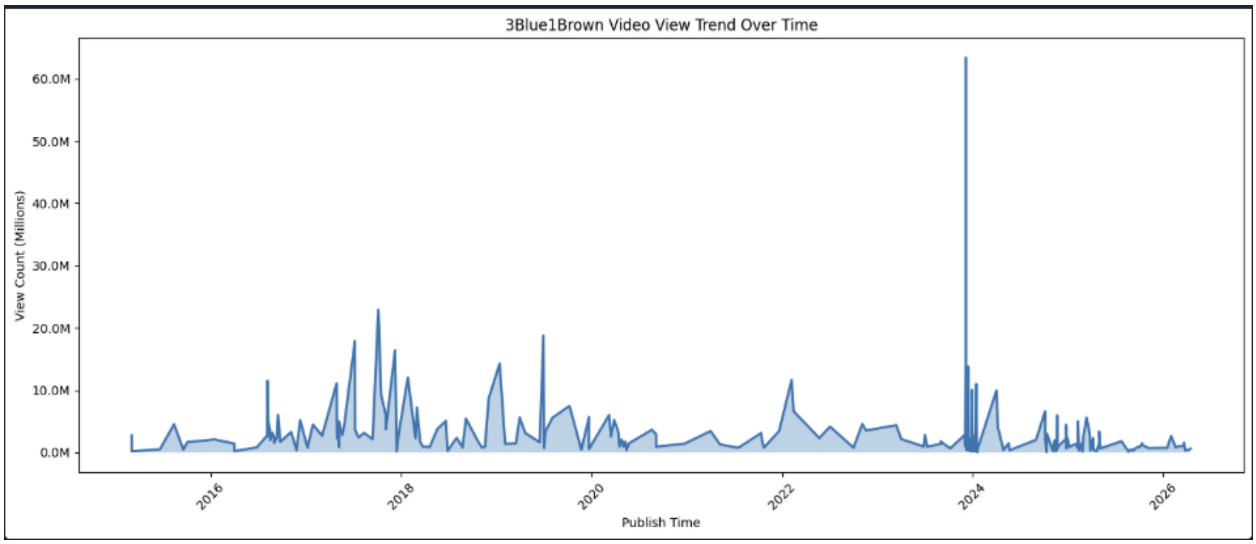
	videoId	title	publishTime	viewCount	commentCount
0	F_0yfvm0UoU	e to the pi i, a nontraditional take (old vers...	2015-03-05 06:15:22+00:00	2751786	2202
1	zLzLxVeqdQg	Euler's Formula Poem	2015-03-05 06:15:23+00:00	180225	162
2	-9OUyo8NFZg	Euler's Formula and Graph Duality	2015-06-21 06:05:43+00:00	518841	386
3	XFDM1ip5HdU	What does it feel like to invent math?	2015-08-14 01:20:50+00:00	4513737	6110
4	1SMmc9gQmHQ	How to count to 1000 on two hands	2015-09-18 21:53:13+00:00	450092	1161

This screenshot shows the extracted dataset that we created from the YouTube API. The outcome proves that we were able to gather the necessary metadata fields and arrange them in

a structured table. Since this dataset has over 200 videos, it provides us with a solid foundation to analyze long-term content, engagement, and sentiment patterns.

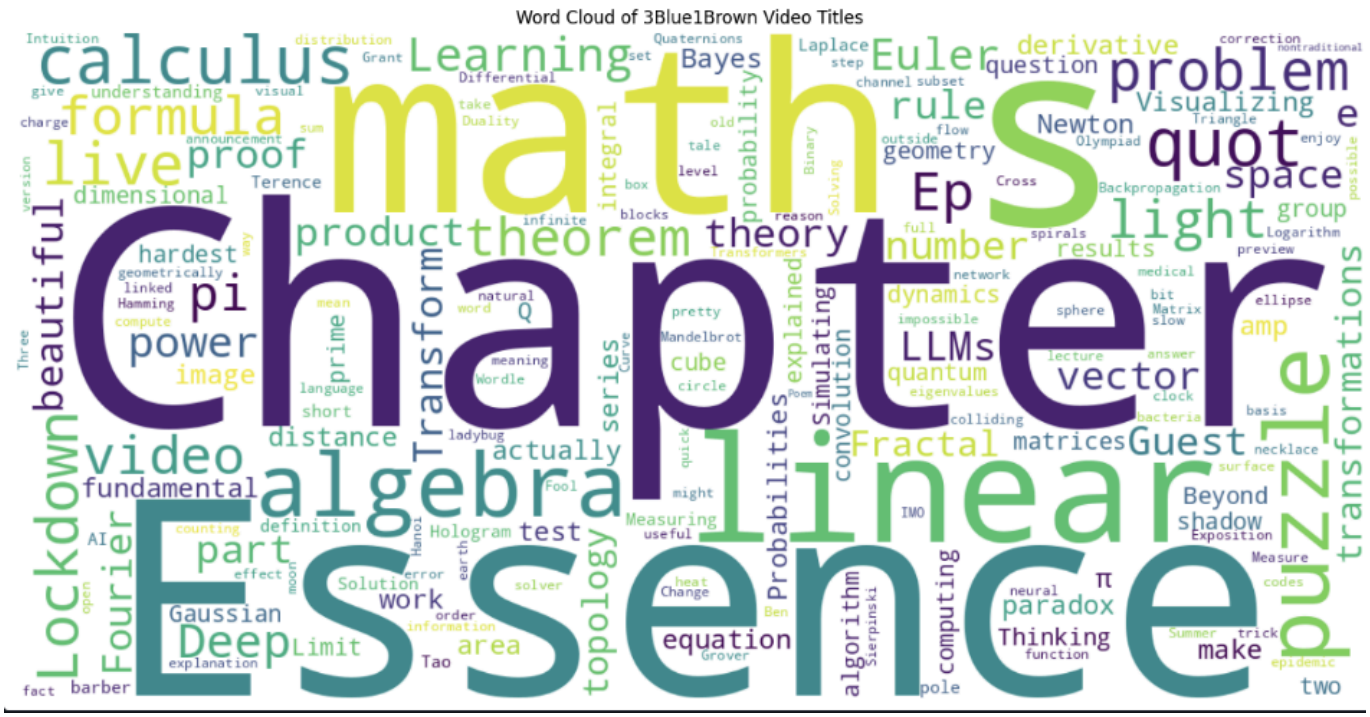
Data Preprocessing and View Trend Analysis

In our view-trend analysis, there are some large outlier videos on the channel that have very large view counts in comparison to the others. The dataset covers videos from 2015-03-05 to 2026-04-16, the highest view count in the dataset was 63,367,491, and the average view count was 3,171,938.78. The majority of videos are well below the maximum spike, indicating that few videos got exceptionally wide coverage over the others on the channel. This trend makes sense to 3Blue1Brown, as certain education subjects become considerably more popular and shared than others. In the following graph, the correlation between the time of video publication and the number of views is presented. One of them is obviously very large spike that exceeds 60 million views, and the rest of the videos are much lower. This informs us that there are some breakout videos in the channel that are far above average. It also demonstrates that not every video has the same attention on the channel as the channel has a long publishing history.



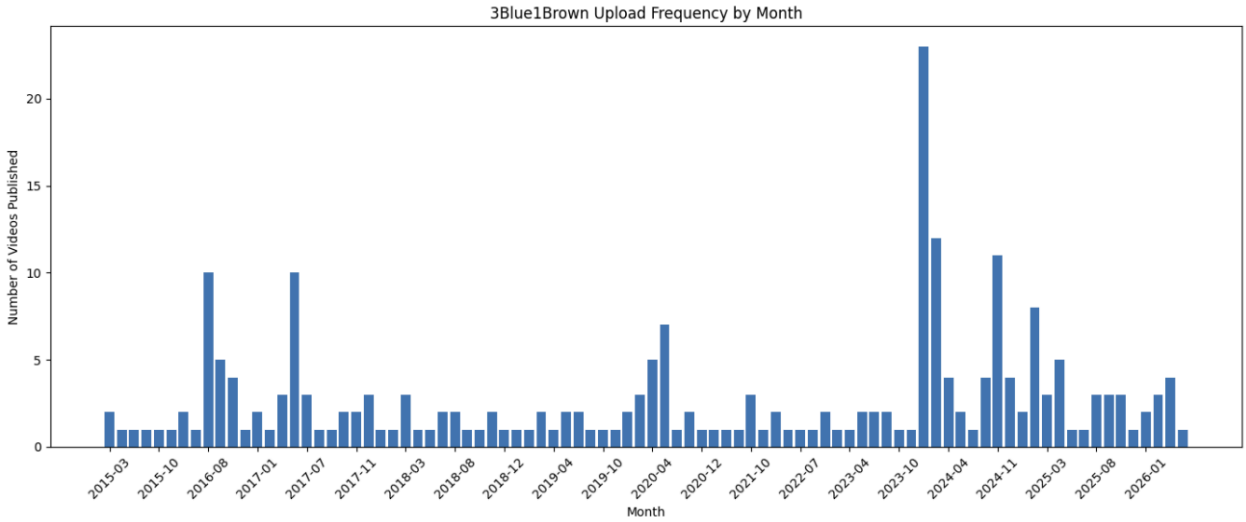
Topic Visualization and Analysis Using Word Cloud

The video title word cloud revealed that 3Blue1Brown has a strong focus on mathematics and conceptual explanation. The most common and biggest words in the visualization were 'Chapter', 'Essence', 'Math', 'Linear' and 'Algebra'. This was confirmed in the notebook output of the most frequently used words, with Chapter being the first followed by Essence, Math and Linear. These words indicate that the channel is much centered on mathematics and educational elucidation. The outcome verifies that 3Blue1Brown possesses an apparent academic identity and revisits similar mathematical subjects in his videos.



Publishing Frequency Analysis

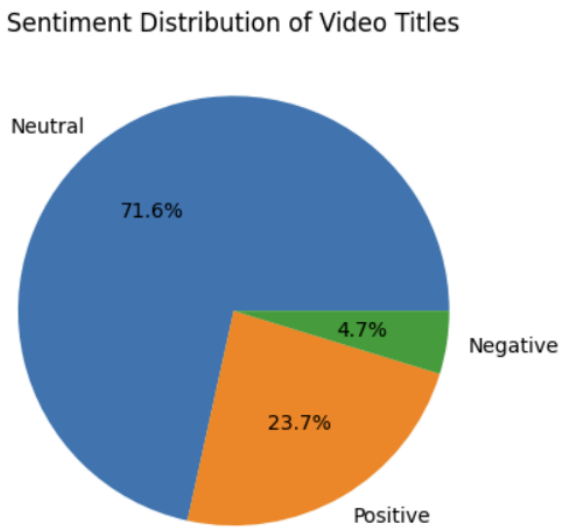
We analyzed our publishing-frequency data to reveal that the channel is not uploading at an ideal constant rate on a monthly basis. Rather, the uploads are in bursts, with some months only having 1 or 2 uploads, and the most active month in the chart with a little over 20 uploads, with the highest number being approximately 23 videos. This trend implies that this channel is probably operating in content cycles, not on a set monthly plan. In the case of an animation-intensive educational channel, this can be explained by the fact that some videos or series can be significantly more time-consuming to prepare and produce than others.



This demonstrates that the channel release pattern is not regular and occurs more in bursts than in a consistent rhythm.

Video Title Sentiment Analysis

The majority of video titles were neutral as indicated by the title sentiment pie chart. In particular, 71.6% of titles were Neutral, 23.7% Positive and 4.7% Negative. This translates to approximately 166 neutral titles, 55 positive titles and 11 negative titles since the dataset has 232 videos. This finding is understandable since educational titles tend to be descriptive and topic-oriented as opposed to being emotional. The prevalence of neutral titles indicates that 3Blue1Brown primarily describes mathematical concepts, ideas, or chapters without exaggeration or opinionated language use.

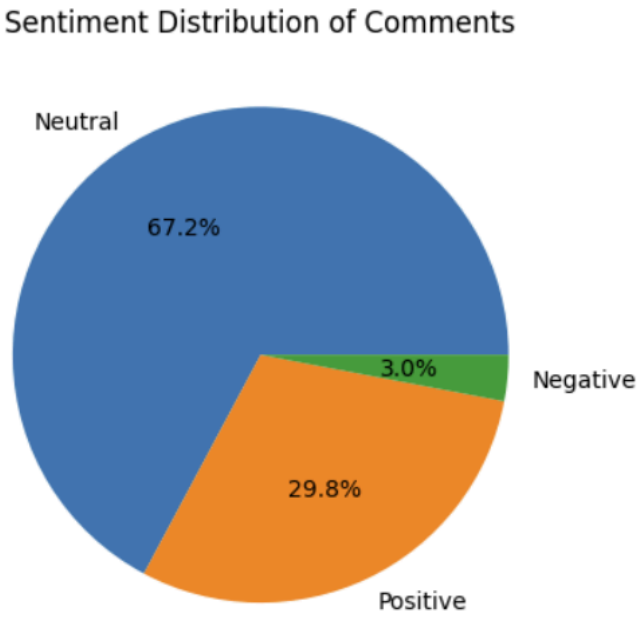


In-depth Comment Analysis

To further analyze the audience, we first determined the top five most-commented videos in the dataset. The video that had the most number of comments was the one titled "But how does bitcoin actually work?" and had 24,303 comments, and the lowest commented video in the top five had only 9153 comments. While he has a respectable amount of audience in his videos, it seems like they do not interact with him by commenting.

	videoId	title	commentCount
46	bBC-nXj3Ng4	But how does bitcoin actually work?	24303
55	OkmNXy7er84	The hardest problem on the hardest test	20274
133	P11ykXwx4-k	I'm still astounded this is true	12070
114	v68zYyaEmEA	Solving Wordle using information theory	9345
58	spUNpyF58BY	But what is the Fourier Transform? A visual i...	9153

We then extracted 500 top-level comments on the chosen video and categorized them as positive, neutral and negative with the help of TextBlob. The findings indicated that 67.2% of the comments were neutral, 29.8% were positive and 3.0% were negative, which translates to approximately 336 neutral comments, 149 positive comments and 15 negative comments. This implies that the majority of viewers were talking about the topic itself, posing questions or making remarks as opposed to publishing highly emotional responses.



The word clouds in the comments provided us with a better perspective of the language employed in each group of the sentiments. The most visible words in the positive word cloud were 'best', 'work', 'great', 'explanation', 'good', and 'understand', indicating that a good number of the viewers liked the clarity of the video.

Positive Comment Word Cloud
best
work
bitcoin
trading
people
great
good
explanation
Mr
explain
now
thing
transaction
know
right
Paulson
Dmitriev
love
say
understand
mining
trade
question
make
profit
will
result
Alice
tech
crypto
well
Expert
need
started
making
kind
first
block
time
used
overnight
strategy
blockchain
cryptocurrency
way
many
host
forex
accurate
wonderful
service
lot
number
videos
success
career
think
getting
better
example
buy
pay
day
consensus
awesome
probably
made
Amazing
Ethereum
believe
honest
real
miners
problem
become
channel
investing
nice
seen
purpose
look
find
broker
reached
learn
math
support
excellent
watched
answer
effort
every
BTC
different
able
start
go
long
grow
trend
last
take
still
someone
output
using
solutions
found
understanding
stick
info
bob
machines
flow
exactly
character
signals
profitable
winning
new
far
wealth
scam
fraud
thought
spend
brilliant
solving
truly
came
hard
farm
chain
Youtuber
invested
glance
investment
Fantastic
send
short
kids
sold
name
notch
tell
subject
income
huge
year
market
together
change
isn
save
length
fact
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excellent
trader
learn
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watched
watched
watched

Conclusion

In this project, we have managed to accomplish the necessary tasks in YouTube data extraction, trend analysis, topic analysis, publishing frequency analysis, title sentiment analysis and in-depth comment analysis. We found that 3Blue1Brown is highly focused on mathematical and educational content with the title vocabulary heavily dominated by the words 'Chapter', 'Essence', 'math', 'linear', and 'algebra'. The trend analysis of views revealed that the channel has several outlier videos with a high performance, the highest being 63,367,491 views, and the mean video views were 3,171,938.78. Analysis of the publishing frequency revealed that the uploading is not a perfectly stable rate, but in bursts. Title sentiment analysis revealed that 71.6% of titles were neutral and comment sentiment analysis revealed that 67.2% of 500 comments analyzed were neutral.

In this project, we learned the fact that real social media data may be noisy is important. The spam, promotional bot-generated content, and repetitive language about cryptos decreased the cleanliness of sentiment-specific word clouds in the chosen comment dataset; we could have worked more on cleaning and normalising the data. This shows a significant big data challenge: gathering information is not the only aspect of the process, cleaning and interpretation of the data are also significant. The same attitude and interaction strategies implemented in this project can be also applied to detect suspicious trends, spamming, and deceptive marketing activity in social media applications. By doing so, the methods can be added to the bigger study of misinformation and poor-quality automated information on the internet.

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3Blue1Brown's Youtube Channel
3Blue1Brown - YouTube. (n.d.). Www.youtube.com. https://www.youtube.com/c/3blue1brown